

THE VIEWER v1.8.0 — R13 trust layer: validate · integrity · sign-off · TM currency · verify cockpit

R13 (above military grade): raise the TRUST, VERIFICATION, and RESILIENCE of the whole app. Additive/rollbackable (R1); append-only audit; no new deps.

1 TRUST & ACCURACY

validate.py -> /api/validate

physical-plausibility + OCR-garble checks
on every extracted value:
quarantine = impossible/garbled (HELD)
suspect = unusual (flagged)
wired into /part: bad data is WITHHELD

trust.py (one canonical level)

high = authoritative + corroborated
medium= single cite review= disagreement
low = external quarantined = failed
validation
so a trust chip means the same everywhere

defense-in-depth accuracy

conflicts.py -> cross-MANUAL disagreement
validate.py -> physical plausibility
two independent checks; disagreement is
SURFACED, never silently resolved.
(cross-method agreement = follow-up)

2 VERIFICATION MADE MEASURABLE

verifystate.py -> /verify (cockpit)

reads the last host-side VERIFY-099 log + the 39-module self-test roster
+ which sidecars are built + DB-integrity status -> ONE 'what have we
proven?' view. Verification you can't see is verification you won't keep green.

tests/test_accuracy.py

MEASURED extraction accuracy vs a hand-verified ground-truth set:
recall / precision with a regression FLOOR. You cannot claim 'above
military grade' without measuring it. Wired into VERIFY-099.

3 HUMAN AUTHORITY & AUDITABILITY

signoff.py -> /review (SME sign-off)

low-confidence value -> review queue -> an expert APPROVES / REJECTS /
OVERRIDES -> verified & locked. The store is APPEND-ONLY: every action is
a new event, nothing is updated or deleted -> a permanent who/what/when
audit trail. extracted -> HUMAN-VERIFIED.

tmrev.py -> /api/tmrev (TM currency)

parses a manual's CHANGE number + date; flags when a NEWER revision of
the same TM exists in the corpus, so no one works from a superseded book.
'Is this the current manual?' answered authoritatively.
change number dominates; date breaks ties.

4 RESILIENCE

integrity.py -> /api/integrity

SQLite corruption detection (integrity_check) · SHA-256 tamper-evidence · a file manifest that catches change/corruption ·
ONLINE-safe backup (a consistent snapshot even while the DB is in use). The index and sidecars are the app's memory —
they must never be silently corrupted or lost. Surfaced in the verification cockpit; complements the off-disk safeguard mirror.

THE POSTURE (R13)

Never shown as fact

A garbled or impossible value is quarantined and withheld — the mechanic sees an honest 'held for review', not a wrong number.

Human vouches

A machine extracts; only an SME verifies. Approvals are permanent and auditable — the record can always be reconstructed.

Proven, not assumed

Every module self-tested; accuracy measured against ground truth; the cockpit makes the proof state visible and keeps it green.

Never lose data

Corruption is detected, tampering is evident, and a consistent backup is one call away. Read-only over the corpus (R6).

R13 above-military-grade (governs all) · R1 rollbackable · R4 CHANGELOG [1.8.0] · R9 VERIFY-099 (6 modules + accuracy) · append-only audit · VERSION=1.8.0.